

The Dynamics of Sand Dunes

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Keywords

sand dunes, sediment transport, dune interactions, dune fields

Abstract

Sand dunes cover 5% of Earth's land surface, and they abundantly populate river bottoms and seabeds. The subtle dynamical interplay between the granular matter and the overlaying fluid leads to rich phenomenology at different scales, from colliding grains through migrating sand dunes to slowly evolving dune fields. In this review, we survey recent developments in the literature on the dynamics of sand dunes and focus in particular on the physics and mathematics. Our discussion is organized around four central paradigms of the field: flat bed instability, single dune migration, dune–dune interactions, and dune field statistics. Besides discussing the key scientific advances, we also highlight the methodological advances in observations, experiments, and simulations that facilitated them. We conclude our review by discussing the social implications of dune dynamics, such as the interaction between dune and infrastructure, and we offer speculation on what research topics related to sand dunes might become important in the next decade.

Dune: bedform with elevated topography

Aeolian: wind-driven

Subaqueous: forming under water

1. INTRODUCTION

It is worth starting our review with the question, What is a sand dune? Etymologically, the word dune is cognate to the word down, and both of them originate from the Germanic word *dūn*, which in Old English meant a low hill. This original meaning of the word down has been preserved in certain geographical names in South England and Australia, for example, the South Downs National Park. In time, the word down became obsolete as a noun, but it has evolved into an adverb describing the descending direction (from a hill toward a lower place). On the other hand, the word dune is still used to describe elevated topography so, as far as the etymology is concerned, a sand dune is nothing more than a sandy hill. The modern use of the word is sometimes more restrictive, and it is based on the understanding that a sand dune is a dynamical structure.

The intricate dynamics of sand dunes has attracted significant attention from several different academic communities, including geographers, engineers, physicists, and applied mathematicians. The earliest book describing dune dynamics, with the evocative title *Waves of Sand and Snow, and the Eddies Which Make Them* underscoring the role of fluid dynamics, was written by Vaughan Cornish in 1914. Another classic resource in this area is a book by Brigadier Ralph Bagnold (1941) of the British Army, who organized and led a range of explorations in the Libyan Desert, was a leading authority on aeolian processes, and published prolifically. As far as the modern literature is concerned, a comprehensive state-of-the-art geological description of sand dune dynamics is presented by Shroder (2022). The course of physical research in sand dune dynamics can be tracked by perusing the topical review articles (Kennedy 1969, Engelund & Fredsoe 1982, Blondeaux 2001, Seminara 2010, Charru et al. 2013, Parteli et al. 2014, Claudin et al. 2024). The aim of this review is to provide a synthesis of the most recent advances in quantitative research on sand dune dynamics, focusing in particular on the findings of the last decade.

The dynamics of a sedimentary bed is tightly related to the dynamics of the overlaying fluid, as sufficiently strong flows have the capacity to transport the grains. This external driving can transform amorphous sediment into distinct self-organized structures, and in this review we generously apply the word dune to most of them (Ashley 1990). The only exception are small-scale aeolian bedforms known as impact ripples, which, as we shall explain later, appear to be coupled to the fluid flow in a unique way. A beautifully illustrated overview of the dune menagerie can be found in the book by Lorenz & Zimbelman (2014). On Earth, dunes can be found not only in deserts (Livingstone et al. 2007, Andreotti et al. 2013) and along the coasts, but also on riverbeds (Best 2005, Coleman & Nikora 2011, Seminara 2010), seabeds (Blondeaux 2001, Vittori & Blondeaux 2024), and even snow fields (Filhol & Sturm 2015). Aeolian sedimentary bedforms are formed by wind-driven processes, while subaqueous bedforms appear in fluvial or marine environments. Sand dunes have also been found on other planets (Mars, Venus, Pluto) (Bishop 2007, Telfer et al. 2018, Day & Zimbelman 2021, Diniega et al. 2022), moons (Titan, Io, Triton) (Bourke et al. 2010, Diniega et al. 2017), and a comet (67P/Churyumov–Gerasimenko) (Jia et al. 2017), where they can be subjected to extreme temperature, density, and/or pressure. In industrial settings, dune-like structures form in multiphase flows, such as cuttings transport (Edelin et al. 2015, Khatibi et al. 2018).

An insightful review of dune taxonomy was recently presented by a collective of authors (Courrech du Pont et al. 2024), who made an attempt to categorize dune nomenclature and vocabulary across fields and identified more than 50 different terms in literature for types of dunes (McKee 1979, Tsoar 1986, Lancaster 1995, Day & Zimbelman 2021). Comparing and contrasting all the different types of dunes is beyond the scope of our review, and our approach is almost the exact opposite. Here, we review the recent advances in fundamental physical research concerned with sand dune dynamics by discussing selected idealized models of reduced complexity.

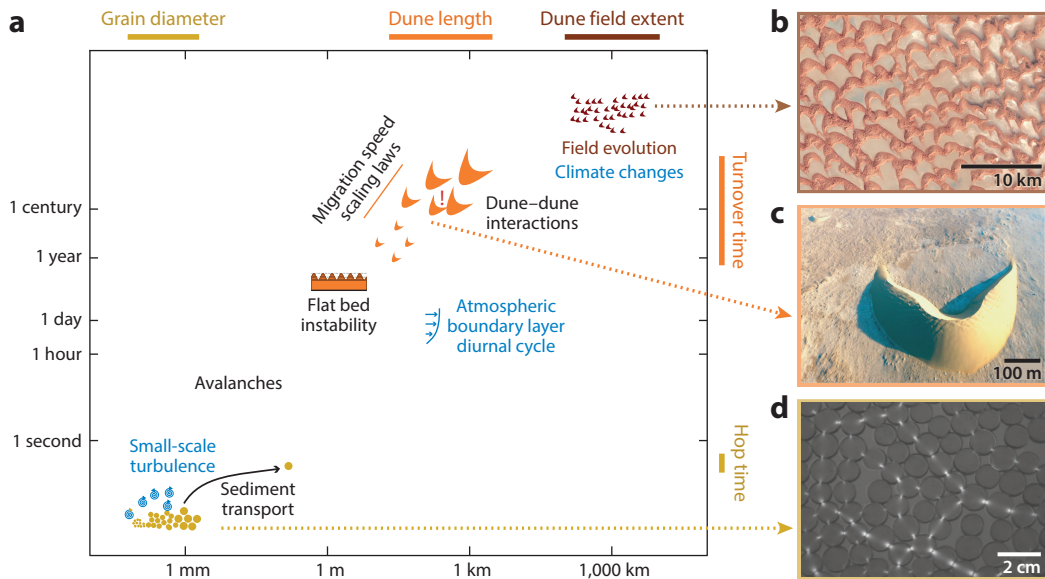


Figure 1

Multiscale dynamics of sand dunes. (a) Diagram illustrating the range of dynamics involved in sand dune mechanics, from microscopic grain processes through mesoscopic dune phenomena to macroscopic evolution of a dune field. (b) Barchan dune field in the Rub-al-Khali Desert on the Arabian Peninsula. Map data copyright 2014 Google, Landsat/Copernicus. (c) Two barchan dunes in Qatar. Image provided by Sylvain Michel. (d) Force chains measured in a photoelastic experiment. Image provided by Benjamin McMillan.

One important challenge in dune modeling is the multiplicity of relevant length scales ranging from individual grains through single dunes to vast dune fields (**Figure 1**). The dynamics of larger/longer scales is a consequence of the dynamics of smaller/faster scales (Werner 1999); for example, small-scale turbulent fluctuations mediate dune–dune interactions, which in turn control the long-term dynamics of a dune field. The physical analysis of sand dune dynamics often relies on the critical insight that small-scale features are salient for the large-scale dynamics.

In this article, we survey a range of research paradigms in sand dune dynamics relating to different length scales. As a starting point, we take the flat bed subjected to a shearing flow that can spontaneously destabilize into a regular undulating pattern. Subsequently, we discuss the dynamics of different types of isolated free dunes, from quasi-2D transverse dunes through 3D barchans to more complex patterns such as star dunes. We then proceed to discussing pairwise dune–dune interactions (first in two dimensions, then in three dimensions), and we finish with a description of the long-term dynamics of a dune field.

2. SEDIMENTARY BED DYNAMICS

2.1. Setup and Scaling

A good starting place to analyze the dynamics of sand dunes is a flat sediment–fluid interface. This geometry can be visualized as a zooming lens view on the surface of a large dune or simply a featureless homogeneous layer of sediment. The five most important parameters of the local dynamics are the diameter of particles d , the density of the particles ρ_p , the density of the fluid ρ_f , the dynamic viscosity of the fluid μ , and the characteristic flow speed u (**Figure 2a**). The precise physical definition of the velocity scale u is a matter of convention; popular choices include

Transverse dune:
straight elongated
bedform

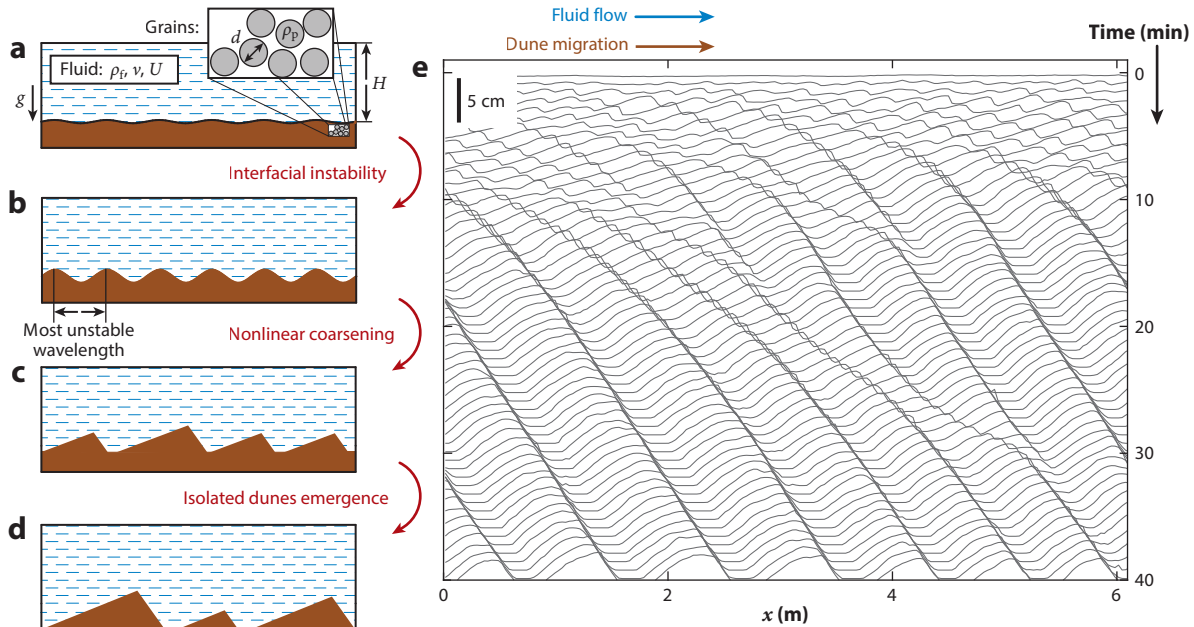


Figure 2

Linear stability and nonlinear coarsening. (a) Early-stage linear stability, growing at the most unstable wavelength λ_1 . (b) Late-stage linear stability, growing at the most unstable wavelength λ_1 . (c) Nonlinear coarsening dynamics, resulting in separating wavelengths. (d) Isolated dune emergence at different wavelengths and different heights. (e) Space-time diagram of position as a function of time, going through the stages shown in panels a–d.

far-field velocity u_∞ , velocity at a fixed height u_z , or shear velocity $u_* = \sqrt{\frac{\tau}{\rho_f}}$, where τ is the shear stress exerted by the flow on the bed. By Buckingham–Pi dimensional analysis, considering that there are three primary dimensions (mass, length, time), there must be three nondimensional groups that control the character of the sedimentary dynamics. The first of them is the particle–fluid density ratio $s = \frac{\rho_p}{\rho_f}$. The second one is the Shields number $\Theta = \frac{\tau}{(\rho_p - \rho_f)gd}$, which compares the magnitude of the two main forces acting on a grain: the hydrodynamic drag and the reduced gravity. There is a freedom of choice for the third nondimensional number, depending on the application. For example, it could be the particle-based Reynolds number $Re_p = \frac{\rho_f u_* d}{\mu}$ based on shear velocity u_* , which represents the ratio between inertial to viscous forces, or the Galileo number $Ga = \frac{1}{\nu} \sqrt{(\frac{\rho_p}{\rho_f} - 1)gd^3}$, which represents the ratio between buoyancy and viscous forces and is sometimes also called the sediment Reynolds number.

2.2. Sediment Transport

The key difference between the two most familiar sedimentary environments (aeolian and subaqueous) is the disparity in particle–fluid density ratio [$s = O(1,000)$ in air and $s = O(1)$ in water]. This difference has important consequences for the momentum–transfer mechanism and grain trajectories. In water, the dominant mechanism of sediment transport is so-called bedload, where grains slide and roll along the interface. This mode of transport is also present in the air, but in aeolian environments we can also observe grains saltating along ballistic trajectories or grains suspended in air. Importantly, as the hydrodynamic drag scales with ρ_f , the subaqueous sedimentary processes operate much faster. This feature is heavily exploited in experimental research,

DUNES IN VITRO

Scaled-down laboratory experiments are essential to capture physical dune behavior and explore the parameter space with controlled conditions. There are a variety of media and geometries that have been used to achieve this goal.

Wind Tunnels

The largest wind tunnel facility, where aeolian as well as Martian bedforms can be reproduced, is the ESA Wind Tunnel in Aarhus, Denmark (**Figure 3a**). The wind tunnel is the only experimental facility that can reproduce impact ripples (Yizhaq et al. 2024). Other types of bedforms are generally easier to study in flume experiments.

Straight Channels

The most popular experimental device in aqueous sediment transport is an inclined open channel, which directly mimics alluvial conditions (Guy et al. 1966). Similar geometries include horizontal flumes with the flow induced by a system of pumps (Groh et al. 2008, Franklin & Charru 2011, Alvarez & Franklin 2017) (**Figure 3b**). The main drawback of all these domains is their finite length, which complicates the boundary conditions and limits the duration of an experiment, but they allow the most precise imaging. Fluid flow around dune-shaped casts has been measured very precisely with particle image velocimetry in an index-matched fluid (Bristow et al. 2018).

Moving Trays

An interesting alternative geometry consists of a small flat tray, which moves through an ambient reservoir (Hersen et al. 2002, Reffet et al. 2010). When the plate reaches the boundary of the reservoir, it can be stopped and slowly pulled back without disturbing the bed (**Figure 3c**). By rotating the moving tray between cycles, one can also use this technique to study the effects of changes in flow direction (Courrech du Pont et al. 2014, Gadal et al. 2019).

Annular Flumes

To circumvent the finite-size limitations, quasi-2D sand dunes can also be created in an annular flume (Betat et al. 1999). This setup is particularly well-suited for long-time investigations of steady states (Bacik et al. 2021b). The use of Taylor–Couette geometry in sedimentary research was pioneered in the 1970s (Engelund 1975). Recent designs place annular flumes on turntables in order to remove some of the rotation-induced effects and facilitate the data capture (**Figure 3d**).

where large slowly evolving aeolian dunes can be emulated with miniature fast-moving subaqueous replicas (see the sidebar titled Dunes In Vitro and **Figure 3**).

The intricate dynamics of sediment transport, involving grain collisions, turbulent intermittency, cohesion, lubrication, and many other factors, is an active field of research, recently rejuvenated by the unprecedented measurement capabilities of direct numerical simulations (Tholen et al. 2023). Nevertheless, not all of the microscopic details impact the macroscopic sediment transport dynamics in an equally significant way. To leading order, the relationship between sediment flux q and shear stress τ can be approximated with simplified transport laws $q(\tau)$, e.g., $q \sim (\tau - \tau_c)^{3/2}$ for water-driven flows and $q \sim (\tau - \tau_c)$ for wind-driven flows, where τ_c is a critical shear stress that signifies onset of sediment transport (Kok et al. 2012).

2.3. Flat Bed Instability

A flat bed of sediment subjected to fluid flow tends to be inherently unstable and can spontaneously transform into an undulating topography. Mathematically, this process can often be explained

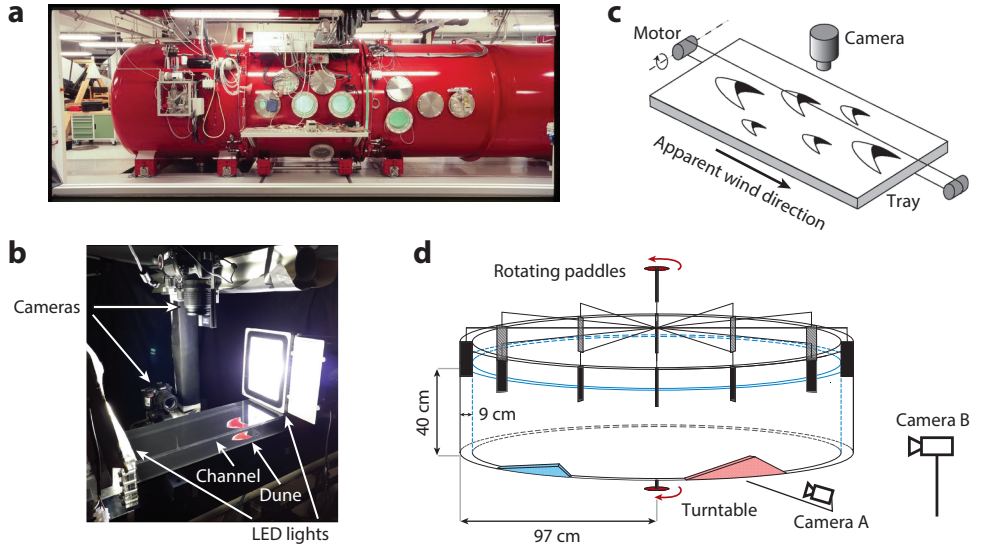


Figure 3

Dunes in vitro. (a) Aarhus wind tunnel. Image provided by the Aarhus University Planetary Environment Facility, Aarhus, Denmark. (b) A water channel with miniature barchan dunes. Panel adapted with permission from Alvarez et al. (2021). (c) A schematic of a moving tray setup. Panel adapted with permission from Hersen et al. (2002). (d) A rotating annulus design suitable for studying long-term evolution. Panel adapted from Bacik et al. (2020).

through linear stability analysis involving three dynamical quantities: shear stress τ , sediment flux q (per unit width), and interface position b .

2.3.1. Classical paradigm. The basic feedback loop can be summarized as follows: Shear stress induces sediment flux, gradients of sediment flux lead to topographical changes, and topographical changes modify the flow. The best understood of these couplings is the relationship between b and q captured in the Exner equation:

$$\frac{\partial b}{\partial t} + \rho^{-1} \nabla q = 0, \quad 1.$$

which follows directly from the conservation of mass, where $\rho = \rho_p \Phi$, and Φ is the granular packing fraction.

To model the effect of shear stress on sediment flux $q(\tau)$, we can use one of the simplified transport laws, described in the paragraph above, together with a relaxation law. The relaxation law draws a distinction between the potential (saturated) sediment flux q_{sat} and the actual (possibly unsaturated) sediment flux q :

$$\frac{\partial q}{\partial x} = L_{\text{sat}}^{-1} (q - q_{\text{sat}}), \quad 2.$$

where L_{sat} is the saturation length. This notion is particularly relevant in aeolian environments, where L_{sat} is of the order of 1 m (Charru et al. 2013). Modeling the coupling between the shear stress and the topography reduces to the problem of fluid flow over an undulating terrain. It is a nontrivial but well-studied problem, which at the linear level amounts to finding an appropriate response operator (Charru et al. 2013).

The key result coming out of the stability calculation is a phase shift between the key dynamical variables. If the sediment flux q has a maximum upstream of the bed topography f , the bedform will grow—sediment will accumulate at the crest and increase and amplify the topography while the bedform moves downstream. This phenomenon, present in both laminar and turbulent flows, has been identified as the key destabilizing mechanism, leading to the classical dune instability.

Crest: the highest point on a dune

2.3.2. The diversity of instabilities: dunes, ripples, megaripples, etc. While the general paradigm of flat bed stability is well-established, the details are still not fully understood. For example, a recent study has revealed an additional mechanism: the hydrodynamic ripple instability, coexisting with the dune instability (Lapôtre et al. 2021, Yizhaq et al. 2024). To rationalize this phenomenon, a new generation of mathematical models that account for turbulent anomaly, bed roughness, and multiple saturation lengths has been proposed (Duran Vinent et al. 2019).

The hydrodynamic ripples have a shorter characteristic wavelength than dunes, but they should not be confused with the centimeter-scale impact ripples, present in the aeolian environments on Earth and on Mars. The impact ripples can be replicated in numerical simulations without the feedback between the topography and the flow (Durán et al. 2014, Yizhaq et al. 2024). Thus, the dynamics of impact ripples is very different from the dynamics of sand dunes, in that it is not so tightly related to hydrodynamics. The dynamics of impact ripples involves resonant movement of grains and screening instabilities (Anderson 1987, Andreotti et al. 2006, Lester et al. 2025) and is beyond the scope of this review.

To accurately describe sand sheet dynamics in different environments, we may need to adjust the idealized linear stability model, taking into account local idiosyncrasies. For example, if the sediment is bidisperse, transport-induced winnowing and segregation (Gao et al. 2016) can lead to the formation of so-called megaripples (Lämmel et al. 2018, Tholen et al. 2022). Moreover, in a multidirectional wind regime, the most unstable direction of the hydrodynamic instability is no longer obvious (Gadal et al. 2019, 2020a), and different bedform dynamics emerge in tidal flows (Vittori & Blondeaux 2022). The nature of the background flow is also very important: Water waves (Van Oyen et al. 2011), free surface interactions in shallow flows (Fourrière et al. 2010, Colombini & Stocchino 2012, Porcile et al. 2023), turbulence (Johnson & Cowen 2020), and many other features can significantly modify the sedimentary dynamics. In summary, the sedimentary environments are too diverse to be analyzed with a one-size-fits-all theoretical framework, and the mathematical modeling of dune dynamics remains a fine art. The task becomes even more challenging in an extraterrestrial context; for example, to describe the giant ripples on comet 67P/Churyumov–Gerasimenko, one has to account for variable gravity (Jia et al. 2017).

2.3.3. Bridging the gap between the theory and experiments/observations. Validating linear stability calculations in experiments and observations can be challenging, not least because the theory implicitly assumes that the perturbations are small, and it is a priori unclear what the limits of validity are when it comes to finite-size undulations. To make a direct comparison, one needs to start with a flat initial condition, as illustrated in **Figure 2a**, for example, by carefully smoothing the sedimentary bed in a wind tunnel (Yizhaq et al. 2024) or an aqueous experiment (Gadal et al. 2019, Jarvis et al. 2022). Impressively, the same experimental protocol has also been replicated successfully in the field by flattening a plot of land in a desert (Delorme et al. 2020; Gadal et al. 2020b; Lü et al. 2021, 2022) (see the sidebar titled Dunes In Vivo and **Figure 4b**). The advantage of laboratory experiments is full control (or at least measuring capability) of the flow conditions, but field-scale experiments are also invaluable in bridging the gap between idealized models and the real world. It turns out that the linear phase (exponential amplitude growth), as illustrated in **Figure 2b**, can last up to a year in the field, and with careful spectral analysis one can extract an empirical dispersion relationship (Lü et al. 2021).

DUNES IN VIVO

Fieldwork is the most classical instrument in sand dune research (**Figure 4a**), but the methodology has significantly advanced since the pioneering observations. Modern field research on sand dunes makes use of the latest technological advances and is not limited to observations but also involves field-scale experiments. Moreover, thanks to remote sensing, satellite images of dunes are now at our fingertips.

Field Measurements

Modern field measurements make use of instruments including terrestrial laser scanning to capture dynamic processes such as avalanches on dunes (Nield et al. 2011), ground penetrating radar to map out internal dune structure and its formation history (Vriend et al. 2012), optically stimulated luminescence dating to determine when grains were last exposed to daylight (Bristow & Duller 2024), and ultrasonic anemometers to measure aerodynamic roughness and shear velocity (Louge et al. 2024). The challenge is not only to capture localized physical behavior on a single dune but also to scale up to an entire dune field. This is usually easier to achieve with remote sensing.

Field-Scale Experiments

By making a topographical intervention in a dune environment, we can also probe dune dynamics in a semicontrolled way. This interesting approach has been used to test flat bed stability theories in both deserts (Delorme et al. 2020; Lü et al. 2021, 2022) and rivers (Fourrière et al. 2010) (**Figure 4b**). This same approach can be used to investigate the efficacy of measures protecting against dune invasion, for example, wooden sand fences or vegetation.

Remote Sensing

Several databases of satellite images, for example, ASTER, SPOT, and Landsat, are now available for academic research. High-fidelity observational data are now also available for extraterrestrial dunes (Day & Kocurek 2018). Such data allow us to track dune field evolution over decades, but the task is ripe in technical challenges, such as the task of dune identification (Emran et al. 2020) (**Figure 4c**). Optical satellite images can also be used as input in the COSI-Corr technique (Leprince et al. 2008) to determine migration rates of dune motion and induced sand flux measurements. By combining validating field measurements with induced sand flux measurements, dune field data can be compared with the sand flux as predicted from the ERA-Interim global circulation model (Michel et al. 2018). Conversely, in planetary environments, dune movement characteristics can be used to infer the nature of the direction and strength of atmospheric flows.

2.4. Beyond the Linear Response

When the bed undulations arising from flat bed instability become sufficiently large, additional nonlinear dynamics ensue. First, we observe a coarsening stage, where the dominant bedform length scale shifts toward larger values, as illustrated in **Figure 2c**. This phenomenon has been robustly observed both in experiments (Betat et al. 1999, Jarvis et al. 2022) and in direct numerical simulations (Zgheib et al. 2018a,b). Some theoretical insights into the coarsening process have been gained from a weakly nonlinear analysis of the governing equations (Hoyle & Woods 1997, Csahók et al. 2000, Yizhaq et al. 2004, Valance 2011), but these models are yet to be confronted with recent experimental data. An interesting consequence of the coarsening process is that once the dunes become longer than the characteristic wavelength of the initial instability, the instability mechanism can recommence; that is, we can observe small secondary undulations developing on top of the larger dunes (Zomer et al. 2021).



Figure 4

Dunes in vivo. (a) Field measurements in Qatar. Image provided by Michel Louge. (b) Landscape-scale experiments in China (Lü et al. 2021). The surface is being flattened in preparation for an instability experiment. Image provided by Clément Narteau. (c) Martian dunes identified with a semiautomatic algorithm. Panel adapted from Emran et al. (2020) (CC BY 4.0).

Eventually, the undulation growth is halted, either by flow depth limitations (Andreotti et al. 2009) or sediment availability (Jarvis et al. 2022, Porcile et al. 2023). In the latter case, the continuous sedimentary bed spontaneously breaks into isolated dunes, as illustrated in **Figure 2d**; these are better described (both ontologically and mathematically) as discrete entities, which we shall discuss next.

3. SINGLE DUNE

In this section, we shift our attention from continuous sedimentary interfaces to isolated heaps of granular matter resting on bare solid substrate, that is, to the paradigmatic sand dunes. A discrete sand dune could be the end product of spontaneous symmetry breaking of a continuous sedimentary bed described in the previous section (Khosronejad & Sotiropoulos 2017, Jarvis et al. 2022, Porcile et al. 2023). Alternatively, it might originate from a growing sedimentary patch or a large amorphous heap known as a protodune (Alvarez & Franklin 2017, Delorme et al. 2023).

We start with a highly idealized, but very didactic, 2D model. Then we discuss different types of 3D dunes ordered by increasing geometric complexity. The most important questions that we ask in this section are (a) what is the shape of the dune? and (b) how fast does it migrate? For simple geometries, our answers are precise and mathematical. For convoluted geometries, our answers are mostly descriptive. Nevertheless, notwithstanding the impressive progress made in the last decades, even the simplest models abound with open questions.

3.1. 2D Slice Model

For unidirectional fluid flow, the sediment dynamics can be decomposed into the transport (streamwise) and the cross-transport (spanwise) directions. As the sand grains move primarily in the transport direction, to first approximation, the sediment transport over a dune can be represented with a 2D model (Andreotti et al. 2002a, Kroy et al. 2002a). Some dunes, for example, so-called transverse dunes, or dunes confined to narrow channels, are indeed spanwise invariant and therefore quasi-2D. However, in principle, any 3D topography can be represented as a stack of quasi-2D slices, so the 2D model is a good place to start analyzing more complex dunes as well.

3.1.1. Elementary topography. The key dynamical variable of the 2D model is the topography $h(x, t)$ describing the shape of the dune. The inherent nature of granular matter places a constraint on the slope that cannot exceed the angle of repose, which for sand equals approximately 30° . In

Angle of repose: critical angle above which granular material collapses under gravity

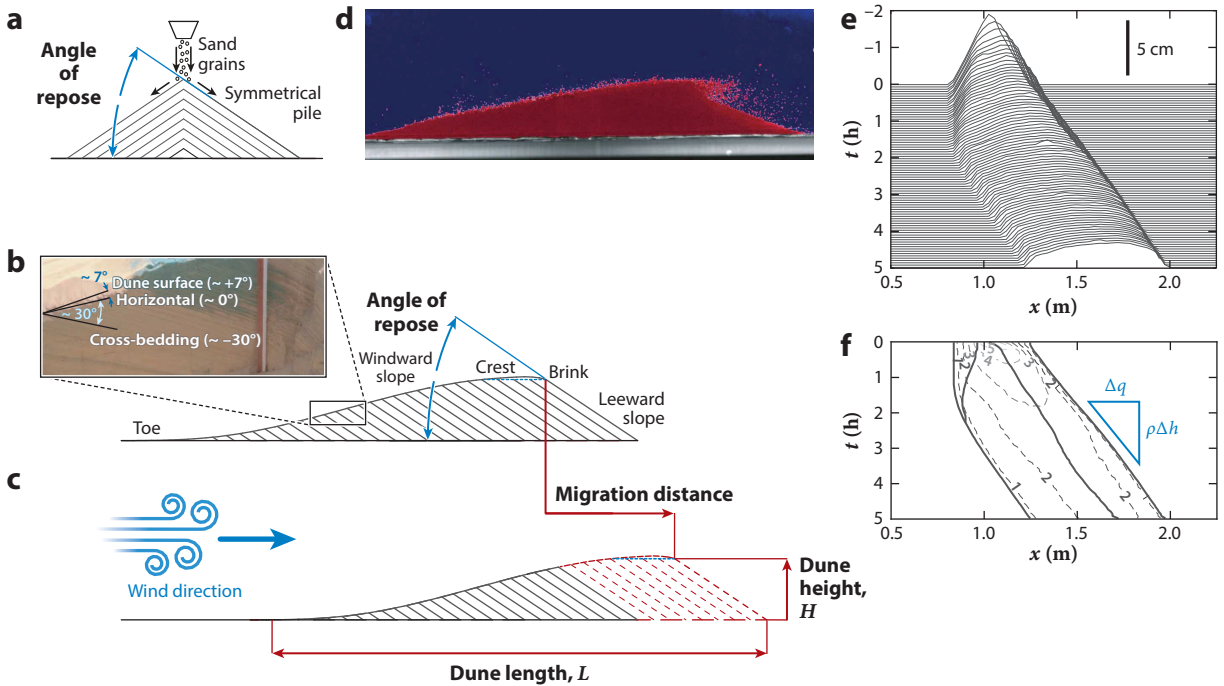


Figure 5

Quasi-2D dune dynamics. (a) Diagram of a symmetrical grain pile at the angle of repose. (b) Diagram of an asymmetrical dune with the leeward face at the angle of repose; the insert shows a dune cross section with cross-bedding at the windward slope, from field experiments in Qatar. (c) Diagram of a migrating dune with evolving cross-bedding. (d) Laboratory subaqueous dune on a periodic racetrack (water is blue, 1-mm diameter ballotini are red), with the sedimenting layer and recirculation zone visible. (e) Space-time diagram of the pile–dune transition recorded in a laboratory racetrack experiment. A quasi-triangular sediment pile ($t = 0$ h) molds into a dune shape ($t = 5$ h). (f) The same data visualized as a contour diagram ($b = 1\text{--}5$ cm) reveal a parallel between dunes and nonlinear waves.

Figure 5a, we show a symmetric quasi-triangular pile of sand with the two sides held at the angle of repose.

In the absence of external forcing, this pile remains consolidated and immobile. However, if we expose it to a sufficiently strong forcing from left to right, it deforms into an asymmetrical 2D dune (**Figure 5b**). Empirically, it appears that for fixed flow boundary conditions, the pile converges to a unique steady dune shape (Groh et al. 2009). To first approximation, the steady sand dune shape remains scale and parameter invariant, except in gentle flows near the motion threshold (Bacik 2021).

The highest point of this shape is known as the dune crest, which is slightly upwind of the brink, defined as where the slope abruptly breaks. This steep downwind, or leeward, slope is still at the angle of repose (approximately 30°) and has been created by grain fall or grain flow processes, as illustrated in **Figure 5b**.

The transformation from a pile to a quasi-2D dune has been studied experimentally, for example, in narrow aqueous channels (Bacik et al. 2020) as shown in **Figure 5e,f**, and with numerical simulations (Narteau et al. 2009). The transition process takes approximately one turnover time, as the pile shifts by its own length (migration distance) and then attains a steady shape. The shape-invariant dune conserves its mass and dimensions (length L and height H), but moves downwind slowly but steadily at a migration speed c , as illustrated in **Figure 5c**.

Brink: the boundary of an avalanching face

3.1.2. Dune migration. It is important to appreciate that a migrating dune does not simply slide along: Its internal structure remains static and the particle transport is confined to a narrow interfacial layer. This thin active sedimentary layer is the only active part of a dune, as clearly visible in **Figure 5d** in a snapshot of an aqueous laboratory experiment. Each individual avalanche leaves behind layers parallel to the angle of repose, not unlike those in a tree-ring structure, which are slowly eroded away as the sand dune moves downstream. Using ground penetrating radar (Vriend et al. 2012, Bristow & Mountney 2022), we can image the internal structure of a dune and see evidence of previous avalanche faces in internal layers, which are now well-preserved inside the dune. The insert of **Figure 5b** shows a photo of the exposed cross section approximately halfway down on the windward slope, where the structure cuts through the dune surface at a slanted inclination. Locally, the dune surface is at $\sim 7^\circ$ from the horizontal, while the cross-bedding is aligned with the angle of repose at $\sim 30^\circ$ from the horizontal.

3.1.3. Sand dune as a kinematic wave. Quantitatively, the heap–dune transition and the steady dune migration can be rationalized by analyzing the Exner equation (Equation 1). In two dimensions, we can rewrite this as

$$\frac{\partial b}{\partial t} + \left[\rho^{-1} \frac{\partial q}{\partial b} \right] \frac{\partial b}{\partial x} = 0, \quad 3.$$

which we identify as the first-order kinematic wave equation with local wave speed

$$c(x, t) = \rho^{-1} \frac{\partial q}{\partial b}. \quad 4.$$

Thus, the evolution of a sand dune can be understood in terms of characteristics, which in a space-time diagram appear as the world lines of constant height (**Figure 5f**) with a slope $c^{-1}(x, t)$. The migration speed of a dune, when it is in steady state, is nearly constant. The dynamics of the downstream edge, the toe of a dune, can be understood as a signaling problem, whereby a nonlinear wave propagates into an unperturbed medium. The initial evolution of the upstream (windward) face can be described as a rarefaction wave with $\frac{\partial c}{\partial x} > 0$. Most interestingly, the crest of the dune could be thought of as a shock front, separating two dynamically distinct regions.

In summary, one can draw a robust mathematical parallel between 2D sand dunes and familiar nonlinear waves, for example, from shallow water dynamics, gas dynamics, or traffic flows. In our opinion, this analogy has been underappreciated in the literature, but we look forward to a more rigorous mathematical treatment of the problem in the future.

3.1.4. Migration speed law. For steady-state shape-invariant dunes, the wave speed c is simply the dune migration rate (illustrated in **Figure 5b,c**). A simple scaling law can be derived from Equation 3 by replacing the derivative with a finite difference:

$$c = \rho^{-1} \frac{\partial q}{\partial b} = \rho^{-1} \frac{q(\text{crest}) - q(\text{toe})}{H}. \quad 5.$$

This migration rate can be measured in experiments, as done in **Figure 5f**, and it is expected to hold for 3D dunes as well. One important caveat is that Equation 5 is not a self-contained predictive law, as the sediment flux is a (potentially complicated) function of the fluid flow around the dune. The flow conditions depend on time and location, and one should not expect that cH is a constant, even within one dune field. In fact, it has been shown that in the absence of direct flux measurements, $c \propto \frac{1}{L}$ (Kroy & Guo 2004), where L is the dune length, or $c \propto \frac{1}{H+H_0}$ (Andreotti et al. 2002a), where H_0 is a constant, gives a more accurate description than $c \propto \frac{1}{H}$.

Cross-bedding: layered structure formed by successive deposition of granular material

Migration rate: the speed at which a dune moves downwind

Barchan dune: falcate (sickle-shaped) migrating dune forming primarily in unidirectional flows

Barchanoid ridge: sinuous elongated bedform; an intermediate structure between a transverse dune and a barchan dune

3.1.5. Flow over the dune. From a hydrodynamic point of view, a sand dune is a wall-attached obstacle perturbing the background flow. As the timescales involved in sand dune evolution are much slower than the timescales for the motion of the overlying fluid, one can assume that the velocity field is at quasi-equilibrium with the topographical boundary condition. The relationship between the velocity field u and the dune topography b can be thought of as a linear input–output problem (see Section 2) (Kroy et al. 2002a, Charru et al. 2013).

While the linearization provides a reasonable approximation at the windward side, it is dramatically inaccurate at the leeward side due to the flow separation near the crest. For typical geophysical flows, the dune-based Reynolds number $Re_H = \frac{\rho c u H}{\mu}$ is almost always high enough to warrant flow separation. The recirculation has important consequences for limiting the outflux of grains, thereby making the dunes approximately self-contained. The characteristics of the recirculation bubble at the downstream end have been extensively investigated numerically (Herrmann et al. 2005, Paarlberg et al. 2007, Araújo et al. 2013), giving a basis for empirical parameterizations used in some of the numerical dune models (see the sidebar titled Dunes In Silico) (Andreotti et al. 2002a,b; Kroy et al. 2002a,b). For inviscid flow, the separation point can also be computed analytically through the Brillouin–Villat condition, giving rise to an elegant analytical model of a migrating sand dune (Charru & Luchini 2019).

3.2. Transverse Dunes

The 2D slice model of a sand dune is easily interpretable as a cross section of a spanwise elongated bedform known as the transverse dune. The dynamics of the transverse dune can be understood by considering a collection of coupled slices exchanging sediment in a diffusive fashion. This dynamics can be described with the so-called migrating string model, where the dune is represented as a curve $x(y, t)$ (the crest line of the dune) whose deformation $\frac{\partial x}{\partial t}$ depends on the gradient $\frac{\partial x}{\partial y}$ (due to avalanches and sediment diffusion), as well as the dune height $b(y, t)$ (due to the migration law) (Niiya et al. 2012, 2013; Guignier et al. 2013). Mathematically, the model reduces to a set of coupled first-order partial differential equations for $x(y, t)$ and $b(y, t)$.

Despite its simplicity, the migrating string model correctly captures two important dynamical phenomena: the transverse dune instability and the falcate shape of the barchan dune. The transverse instability is a spontaneous mechanism whereby a straight transverse dune [$x(y, t) = x_0$] spontaneously deforms into a sinuous bedform known as the barchanoid ridge (Parteli et al. 2011). This instability occurs when the destabilizing downstream lateral transport prevails over the stabilizing upstream lateral transport (Guignier et al. 2013).

3.3. Barchan Dunes

The barchan dune is one of the most common dune types on Earth. Barchan dunes are crescent-shaped dunes with a steep downwind slip face scattered on a substrate of gravel and granules. These dunes with irregular spacing and size exist in sand transport corridors and are characterized by a unidirectional wind regime, typically with variations within 15° of the mean direction (Tsoar 1986), and a low availability of sand. The width of the dune is roughly 10 times the dune height and roughly equal to the dune length.

To first approximation, the crest line of a steady-state barchan dune can be described as a shifting parabola. The parabolic profile is indeed a steady-state solution within the migrating string model described in the previous section (Guignier et al. 2013). The bow-shaped profile can also be obtained as a numerical solution to a 3D extension (Schwämmle & Herrmann 2003, 2005a) of the 2D minimal dune model (Kroy et al. 2002b) (see the sidebar titled Dunes In Silico). Nevertheless, barchan dune dynamics is also rich in details that cannot be captured by these idealized models.

DUNES IN SILICO

Numerical simulations are an increasingly important research tool in sand dune dynamics. The techniques range from exact digital twins, which resolve individual grains and directly simulate the fluid flow, to semianalytic models of specific bedforms.

Grain-Resolved Simulations

The two building blocks of grain-based simulations are a direct computational fluid dynamics (CFD) simulation of the flow and a discrete element method (DEM) simulation of individual grains. The two are coupled through an empirical drag force law. CFD-DEM simulations are versatile and accurate, but they are computationally expensive. They have been used to investigate the microscopic properties of sediment transport (Pähtz et al. 2020) and to simulate isolated (Lima et al. 2022) and interacting (Lima et al. 2024) barchan dunes.

Eulerian Solvers

A computationally cheaper, yet still highly accurate, alternative is to couple the full CFD simulation with an Eulerian interface through one of the empirical $\tau - q$ closures (Khosronejad & Sotiropoulos 2017; Zgheib et al. 2018a,b).

Cellular Automata

An interesting heuristic simulation package called ReSCAL was developed by Narteau et al. (2009) and Rozier et al. (2019). Its fluid solver is based on a lattice Boltzmann method, and the bed evolution is approximated with a cellular automaton. Although it cannot resolve the details of turbulence with the same accuracy as CFD models, it is less expensive computationally and equally versatile. It has been used to simulate interacting 2D dunes (Jarvis et al. 2022), linear dunes (Rozier et al. 2019), and star dunes (Zhang et al. 2012).

Minimal Models

For unidirectional flows, heuristic flow models can be used to reduce the dimensionality of the fluid problem and to construct a semianalytic model of a migrating sand dune in either two dimensions (Andreotti et al. 2002a,b; Kroy et al. 2002a,b; Charru & Luchini 2019) or three dimensions (Durán et al. 2010). The dynamical variables of these models are dune topography $b(x, t)$, sediment flux $q(x, t)$, and shear stress $\tau(x, t)$. The main component of the model is a set of partial differential equations coupling these variables, but a closure is required to model flow separation and avalanching. Although these models are constrained to specific situations, they are fast and interpretable.

3.3.1. Grain motion across a barchan dune. Sand is eroded from the windward face due to saltation and creep and deposited on the leeward face due to grain flow in avalanches and grain fall due to loss of momentum in the recirculation bubble. Some significant sideways migration of grains may exist in these 3D structures: A grain on the midline will be recaptured by the leeward slope and reinserted into the barchan system, while grains migrating to the horns could leave the system by erosion (Alvarez & Franklin 2018, 2019).

3.3.2. Flow around a barchan. The motion of grains across a barchan is induced by the drag forces due to the turbulent flow of the overlying fluid (Alvarez & Franklin 2021). Different facets of this highly complex flow (wake structure, hairpin vortices, etc.) have been the subject of detailed experimental (Charru & Franklin 2012, Palmer et al. 2012, Bristow et al. 2020) and numerical (Omidyeganeh & Piomelli 2013a,b; Wang & Anderson 2018) studies.

Calving: spontaneous separation of small barchan dunes at the flanks of a large barchan dune

3.3.3. Barchan instabilities. Barchan dunes are remarkably robust self-organized structures, but under the right conditions they can also lose their integrity. For example, under bimodal winds, the horns can elongate in an asymmetric manner (Parteli et al. 2014, Lü et al. 2016). As we discuss in Section 4, a similar effect can also be induced by the presence of another dune. The horns of a dune can also undergo so-called calving, which leads to the generation of small new dunes (Worman et al. 2013).

3.4. Complex Dunes

According to Curie's symmetry principle, the symmetry of the causes determines the symmetry of the effects. The primary cause of the dune dynamics is the forcing, and its symmetry indeed determines the morphology and the dynamics of the dune. So far we have discussed dunes immersed in unidirectional flows, but sand dunes can also be found in oscillatory flows, bidirectional flows, and multidirectional flows.

3.4.1. Oscillatory flows. While in some geometries the flow is unidirectional (e.g., riverbeds) or unimodal (e.g., winds with one dominant direction), there are many natural situations where the flow switches between two directions. As a consequence, bedform dynamics in these locations is spatiotemporally symmetric. Moreover, as the switching timescale is typically much shorter than the formation time of a dune, the topography often remains spatially symmetric and time invariant. A classic example of such a scenario is given by coastal ripples formed in tidal flows (Blondeaux 2001). Flow reversals can also reset and rebuild existing dunes (Assis et al. 2025), and dune reversal times are larger than the initial heap-to-dune formation time. Oscillatory ripples are similar to linear dunes, but their cross section is inevitably symmetric. The reflection symmetry also prevents them from transforming into barchanoid ridges.

3.4.2. Bidirectional flows. In aeolian systems, the two wind directions do not necessarily need to be parallel, but they can be determined by the local climate. In the bidirectional wind regime, the dominant wavelength of a flat bed instability can be oblique to both winds (Gadal et al. 2019, 2020a). A coherent heap of sediment does not morph into a barchan dune, but it elongates into a long linear dune (Lucas et al. 2015, Rozier et al. 2019). Interestingly, the orientation of the linear dune depends not only on the wind distribution but also on the sediment availability (Gao et al. 2015, Fernandez-Cascales et al. 2018). If the wind switching timescale is sufficiently long, such a dune can also exhibit seasonal reversals of migration (Gao et al. 2021). A linear dune in a bidirectional flow can also deform into a raked linear dune due to a superimposed surface instability (Lü et al. 2017).

3.4.3. Multidirectional flows. Even though such cases are more rare in nature, theoretically we can also consider tridirectional flows (Rubin et al. 2024) or more. Each of these cases can lead to the formation of new distinct bedforms with distinct symmetries. For example, in multidirectional flows, we observe the formation of star dunes (Zhang et al. 2012). Complex dunes have received significantly less attention in the physical community than barchan dunes; for example, we are not aware of any studies of the flow around star dunes, so they offer rich opportunities for future research.

3.5. Dunes in Confined Geometries

So far we have implicitly assumed that the dune rests on a plane and its size is limited only by the sediment availability. However, in many practical scenarios, the dune dynamics is constrained by the surrounding geometry. The most obvious example is given by dunes in pipes (Edelin et al. 2015) and channels (Zgheib & Balachandar 2019), where the size is limited by the boundaries. In

such cases, we can also observe distinct types of bedforms, such as alternating bars, or upstream-migrating antidunes.

4. DUNE–DUNE INTERACTIONS

Sand dunes rarely occur in isolation, but they almost always interact with their surroundings, and most importantly with other dunes. R.A. Bagnold (1941, p. 214) asserts in his classical text *The Physics of Blown Sand and Desert Dunes* that “in nature we nearly always find that the complication in the dune outline is due to the presence of another dune immediately to windward.” Photographic evidence to support this statement, including time-resolved images, became available with the advent of satellite imaging. Observational studies show that when dunes are sufficiently close, the shape of the downstream dune indeed starts being affected. Surprisingly, the dunes do not always amalgamate into one structure, but they can maintain their discrete identities throughout the interaction, as illustrated in **Figure 6**. In a sense, this behavior of dunes is not dissimilar to a collision of particles or solitary waves, which come close together and bounce off each other (Hersen & Douady 2005, Vermeesch 2011, Hugenholtz et al. 2012). These intriguing findings motivate experimental and numerical research, where dune–dune interactions can be replicated and investigated in a controlled environment. We now describe them by starting from a simplified quasi-2D setting.

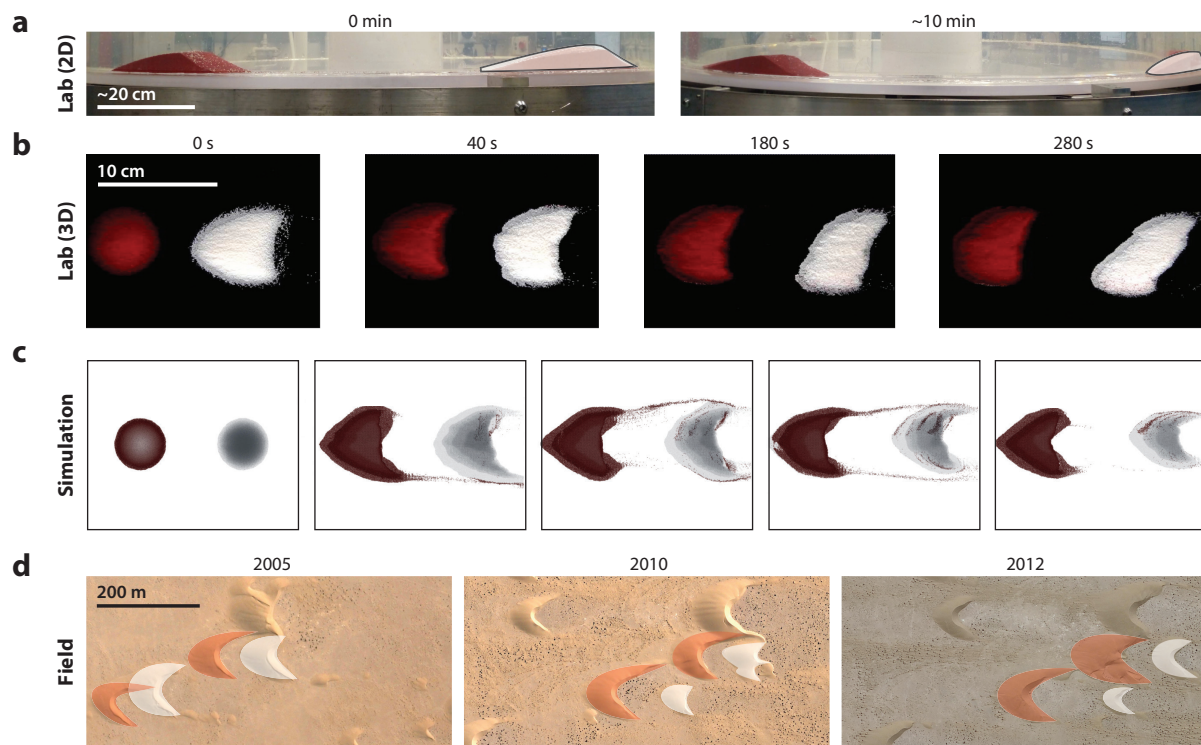


Figure 6

Dune–dune interactions. (a) Repulsion of quasi-2D underwater dunes in an annular experiment. Panel adapted from concepts presented in Bacik et al. (2020). (b) Repulsion (chasing) of underwater barchan dunes in a controlled experiment. Panel reproduced with permission from Assis & Franklin (2020). (c) Direct simulations of interacting barchan dunes. Panel reproduced with permission from Lima et al. (2024). (d) Dune field in the western Sahara, with two highlighted pairs of interacting dunes. Map data copyright 2012 Google, Maxar Technologies.

Dune–dune

repulsion: long-range wake-induced interaction between two dunes preventing a collision

Ejection:

redistribution of sediment between two colliding dunes

4.1. 2D Interactions

As discussed in Section 4, steady-state dunes migrate with speed inversely proportional to their height. One could naïvely hypothesize, therefore, that two quasi-2D dunes of equal volume immersed in the same stream would migrate at the same rate, and two dunes of different volume would either separate or collide, depending on their relative configuration. This reasoning is too simplistic because it does not account for hydrodynamic interactions or sediment exchange between the two dunes. Recently, this phenomenon was demonstrated in an experiment with two dunes of equal volume confined to a narrow annular conduit (Bacik et al. 2020). The downstream dune in this setup is markedly faster due to turbulence-induced enhancement of sediment flux, and as a result the dune–dune separation continues to increase (**Figure 6a**). The turbulent structures impinging on the downstream dune originate from the wake of the upstream dune (Katsuki et al. 2011), whose structure is a rich subject of its own (Dey et al. 2020). This phenomenology has also recently been replicated by a 2D cellular automaton model (Vérité et al. 2025).

While dune–dune repulsion can effectively prevent collisions of dunes similar in size, for sufficiently different volumes, 2D dunes can come in physical contact. When this happens, dunes either amalgamate into one larger structure or redistribute mass through a process known as ejection (Diniega et al. 2010, Jarvis et al. 2022).

4.2. 3D Interactions

Dune–dune interactions also occur for 3D dunes, and they can be substantially more complex. The most well-studied are the interactions of barchan dunes.

4.2.1. Experiments and simulations. Interactions of barchan dunes were replicated in subaqueous experiments almost two decades ago (Endo et al. 2004, Hersen & Douady 2005) (**Figure 6b**). Nevertheless, barchan dune interactions are more complex, not least due to the additional geometric parameter–spanwise offset. Thus, we had to wait several years for the classification of all possible outcomes of a barchan–barchan interaction, including chasing, splitting, amalgamation, etc., to be completed (Assis & Franklin 2020, 2021; He et al. 2023). Modern experimental methods also track the motion of individual grains during binary barchan collisions, including trajectories and origin and destination of grains during different interaction patterns for both monodisperse (Assis & Franklin 2021) and bidisperse sands (Assis et al. 2022).

Certain types of barchan–barchan interactions can also be replicated by the 3D extension of the minimal model (see the sidebar titled *Dunes In Silico*) (Durán et al. 2005, Schwämmle & Herrman 2005b). A new generation of direct numerical simulations allow the reproduction of dune–dune interactions at the grain scale (Lima et al. 2024) (**Figure 6c**).

4.2.2. Wake structure. Direct numerical simulations of interacting dunes use a turbulent closure model (e.g., large eddy simulations) to simulate the flow, which appears to capture the dominant interaction effects well. More detailed measurements of the turbulent structures in the wake have been performed with index-matched fluid experiments (Bristow et al. 2018). However, as the sand dunes are replaced with casts, such studies only permit investigating the flow structure for selected stages of the interaction.

5. DUNE FIELDS

Sand dunes often form large collectives, known as dune fields or dune corridors. At this level of resolution, we are no longer concerned with the particulars of individual dunes, but with averaged quantities, such as dune density or mean migration rate. The field-scale perspective offers an appropriate viewpoint for describing a range of emergent geological phenomena. For example, we

can study how the pairwise dune–dune interactions control the long-term evolution of a sandy desert. One particularly pressing task is to understand how these dynamics might be affected by the changing climate. Conversely, we can also explore how the presence of a dune field influences atmospheric flows.

5.1. Collective Dynamics

When we look at a satellite image of a dense dune field we can easily understand the enthusiasm of R.A. Bagnold, who wrote, “The observer never fails to be amazed at a simplicity of form, an exactitude of repetition and a geometric order unknown in nature on a larger scale than that of a crystalline structure” (Bagnold 1941, p. xxi).

5.1.1. Assessing regularity: order parameter analysis. Conscious of the human tendency for perceiving meaningful patterns, we should not rely on visual assessment alone. Instead, the regularity of a dune field should be evaluated with a statistical order parameter.

Statistical studies of dune field organization are surprisingly scarce, but we can find some notable exceptions (Ewing & Kocurek 2010, Day & Kocurek 2018), including studies dedicated to planetary bedforms (Marvin et al. 2023). For example, nearest neighbor (NN) analysis (Hugenholtz et al. 2012) has confirmed statistically significant regularity of several terrestrial and extraterrestrial dune fields by comparing the observed distribution of dunes to a statistical null model (Bishop 2007, 2010, 2013). Even more compelling evidence in support of the alleged self-organization of dune fields (Andreotti et al. 2002a, Hersen 2004, Kocurek & Ewing 2005) was provided by Wilkins & Ford (2007). The authors of this study tracked the changes of the Coral Pink Sand Dunes in Kane County, Utah, between 1960 and 1997 and found that the NN-based order parameter significantly increased in that period. Hopefully, with the advancement of remote sensing (Baird et al. 2019), in a few decades we will have enough time-resolved data to perform similar analysis of other dune fields as well.

5.1.2. Explaining regularity: agent-based modeling. The spontaneous increase in dune field regularity tacitly implies the existence of a self-organizing principle. The self-organization hypothesis can also be demonstrated explicitly with an agent-based (AB) mathematical model. In this approach, dunes are represented as discrete particle-like objects, moving and interacting according to simple empirically constrained rules. In two dimensions, the reduced-order description of a migrating dune is comprised of just two numbers (position x and size b), coupled through the inverse height law $\dot{x} \propto H^{-1}$ (see Equation 5) (Lee et al. 2005, Ning et al. 2019). By parameterizing wake-induced (Bacik et al. 2021b) and collision-like interactions (Diniaga et al. 2010), we can close the system and simulate the dynamics of a multidune system. Even in this simple setting we can find surprising phenomena, such as oscillations or instabilities (Bacik et al. 2021b). The system becomes more complex for barchan dunes moving on a plane. The most recent generation of AB models describe barchan dunes as a 4D vector (x, y, r, l) with two position dimensions and two shape dimensions to allow for spanwise asymmetry (Robson & Baas 2023). This is different from the previous models that described the dune size with just one number (Lima et al. 2002; Baas 2007; Durán et al. 2011; Eastwood et al. 2011; Bo & Zheng 2013; Génois et al. 2013, 2016; Worman et al. 2013). The inclusion of the additional shape dimension improved the parsimony of the interaction encoding, which needs to capture calving, aggregation, fragmentation, mass exchange, etc. The main motivation behind AB modeling is to predict the long-term evolution of a dune field. An appropriately calibrated model can yield steady-state size distributions in barchan swarms, which for sufficiently simple models can also be computed analytically (Robson et al. 2021).

Giant dune: very large dune; can reach several kilometers in length on Earth

Shadow dune: immobile sedimentary deposit downstream of an obstacle

5.2. Field-Scale Hydrodynamics

The dynamics of a dune field is coupled to the dynamics of the atmospheric boundary layer (ABL) (Bristow 2022). For example, the daily heat cycle can induce a circadian periodicity in dune migration, with peak activity in the afternoon (Gunn et al. 2021). It has also been proposed that the height of the ABL controls the height of giant dunes (Andreotti et al. 2009), that is, that topography–flow interactions prevent dune growth beyond a certain critical height. However, this theory has recently been challenged (Gunn et al. 2022). The competing hypothesis surmises that unless limited by sand supply (Jarvis et al. 2022), sand dunes keep growing at a logarithmic rate.

Sand dunes are not passively driven by the wind, but they also affect the large-scale structure of the ABL. Indeed, each dune imposes a form drag on the incident wind. As a result, turbulent flow through a dune field can be understood as a flow over a rough surface (Gadal et al. 2022, Jia et al. 2023). For example, if a dune field is adjacent to a smooth playa, the abrupt increase in roughness at the upwind margins of the dune field triggers the development of an internal boundary layer that thickens downwind and causes a spatial decrease in the surface wind stress, with far-reaching consequences for the landscape dynamics (Jerolmack et al. 2012, Gunn et al. 2020).

6. DISCUSSION

6.1. Significance

Sand dune dynamics is not only an interesting object of study, but it also has tangible implications for industries and local communities.

6.1.1. Dune encroachment. One of the main hazards that arises as a consequence of dune migration is dune encroachment on human-made infrastructure erected in sedimentary environments (Andreotti et al. 2013, Lorenz & Zimbelman 2014). In aeolian environments, sand dunes notoriously bury roads, rail tracks, and even buildings (Bruno et al. 2018, He et al. 2018). One of the most famous dune encroachment locations is the Mauritanian capital, Nouakchott, which is surrounded by active dunes (Gómez et al. 2018). Dune encroachment also leads to a degradation of arable land, which is why dune fixation for decades has been investigated and supported by the Food and Agriculture Organization of the United Nations (Berte 2010). Dune fixation can be achieved by biological (planting vegetation), chemical (petroleum products spread over the surface), or mechanical means (protective screens and mulches). Dune motion can also be controlled with fences, which block or divert sediment flux, or topographical streamlining, which allows the sediment to be swept away more quickly. Large dunes can also be removed from desert environments in large-scale operations involving bulldozers and hauling trucks; an example can be observed in Google Earth in January 2016 at latitude N 25°0′40.09″, longitude E 51°23′55.16″, with finely grained sand to be used in construction applications. The migration of underwater dunes is of practical interest as well, as it entails the dredging of navigation channels and compromises the safety of subsea cables and pipelines (Barnard et al. 2006).

The physics of dune–obstacle interaction has been recently investigated in idealized subaqueous laboratory experiments, both for quasi-2D dunes in a narrow channel (Bacik et al. 2021a) and for 3D barchan dunes (Assis et al. 2023). Depending on the obstacle shape and size, as well as the sediment properties, the dune can either cross the obstacle and recover as a migrating dune or get trapped in the wake of the obstacle and turn into a shadow dune. It has also been shown that for 2D dunes, the outcome of the dune–obstacle interaction can be predicted by analyzing the structure of the fluid flow around a given obstacle (Bacik et al. 2021a). A similar analysis for more realistic 3D dunes is yet to be performed.

6.1.2. Sand dunes and the climate. A question of current importance is to what extent aeolian dune dynamics can be affected by climate change. The dynamics of aeolian sand dunes is primarily driven by the winds, so changes in atmospheric flows will undoubtedly affect dune migration. An increase in wind velocity can not only increase the migration rate of active dunes but also remobilize currently immobile dunes (Thomas et al. 2005). A change in wind circulation patterns can induce a transition in dune orientation or dune type (e.g., from an asymmetric dune driven by a unidirectional flow to a symmetric dune driven by a bidirectional flow) (Baas & Delobel 2022).

With dune fields covering approximately 5% of Earth's land surface (Thomas et al. 2005), changes in sand dune dynamics can not only affect the local environments, but also feed back to the climate dynamics, for example, through increased dust emission (Harrison et al. 2001) or vegetation–albedo feedback (dark vegetated immobile dunes versus high-albedo mobile dunes) (Yizhaq et al. 2009, Zeng & Yoon 2009).

6.2. Research Perspectives

The oldest tool in our arsenal to capture and investigate sand dune dynamics is to use observations. Marco Polo [2006 (1295)] described the singing sand dunes along the Silk Road in the 13th century, which “at times fill the air with the sounds of all kinds of musical instruments, and also of drums and the clash of arms.” The loud and thundering rumbles coming from the dunes were the work not of spirits but of physics. Later, Vaughan Cornish (1914, p. 53), writing in Victorian language, eloquently expressed dune–dune interaction as “one [dune] behind the other, each dependent in form, position, and movement upon . . . that to windward.” Modern sedimentary geomorphology started with the seminal work of Brigadier Bagnold (1941), who explored the Libyan Desert during his military campaigns, built a wind tunnel upon his return to England, and published a field-defining book titled *The Physics of Blown Sand and Desert Dunes*, which is still used as a reference today. Observations and field measurements are still powerful but also challenging, as logistical constraints lead to limited data (see the sidebar titled Dunes In Vivo and **Figure 4**).

As our research tools evolve, well-controlled experiments in either aqueous flumes (Bacik et al. 2020, Jarvis et al. 2022) or wind tunnels (Rasmussen et al. 2011), equipped with high-resolution instrumentation, are forming the state of the art (see the sidebar titled Dunes In Vitro and **Figure 3**). Physical scalings and measurements in analog experiments are essential to capture the essence of sediment transport and bedform dynamics. There is also a strong role for mathematical modeling, as idealized experiments can be used to validate mathematical models, whether computational (Kroy et al. 2002b) or analytical (Cannarsa & Vita 2024).

Remote sensing has also improved in recent years, and optical satellite images can be processed with orthorectification and coregistration to create horizontal ground displacements of dune-scale areas by correlation (Leprince et al. 2008). Numerical simulations are getting more accurate and realistic with increasing computational power (see the sidebar titled Dunes In Silico). We will never be able to simulate the particle–particle interaction on a dune or field scale, but we are able to make the transition across scales and mimic the continuum behavior better (Werner 1999).

What might be covered in the next *Annual Review of Fluid Mechanics* article on the dynamics of sand dunes in a decade or two? One important subject would be to clarify the connections between different flat bed instability mechanisms (Yizhaq et al. 2024, Lester et al. 2025); machine learning could potentially play a critical role in this debate. Indeed, we feel that with the arrival of machine learning and artificial intelligence (AI), we might see a major shift in methodology. While AI algorithms are already being used to aid image analysis (Cúñez & Franklin 2024), we can also see applications of system identification techniques (Brunton et al. 2020, Scheiber et al. 2024) to extract data-driven models. One particularly important challenge that such models might help us with is forecasting the impact of climatic changes on dune environments and its consequences for society.

SUMMARY POINTS

1. Sand dune dynamics can be analyzed at different levels of resolution, from grains through individual bedforms to entire dune fields. Large-scale dynamics subsists on small-scale dynamics, but not all the microscopic details are equally relevant.
2. The complex dynamics of dunes can be efficaciously disentangled by studying stereotyped models (e.g., a flat interface, a quasi-2D sand dune, or an isolate barchan), even if the idealizations do not capture the full complexity and diversity of natural dunes.
3. The interactions between individual dunes are both complex and fascinating, depend on geometry and forcing due to wake-induced feedback, and lead to an intricate palette of possible outcomes, including repulsion, collision, splitting, etc.
4. Dune fields and corridors feature a range of scales, can be forecast with an agent-based model, and are impacted by climate change and the atmospheric boundary layer.
5. Dune encroachment, the interaction between nature and humans, and the changing climate are dynamic parameters that will change the face of deserts and arid regions.

FUTURE ISSUES

1. In sand dune literature, two ongoing debates are particularly active: one in quest of a unified model of flat bed instability and another one regarding the limiting role of the atmospheric boundary layer in giant dune formation. Key advances in both areas are likely to come in the next decade.
2. Despite significant progress, sediment transport and sedimentary layer instabilities are still active areas of research. Thanks to direct numerical simulations, we have recently started to appreciate the microscopic details of granular mechanics of the bed. The significance of small-scale turbulence on sediment transport and flat bed stability is yet to be charted. Connecting to this is ongoing research regarding the time-dependent signatures of the onset of sediment transport.
3. Novel machine learning tools, such as nonlinear system identification and physics-aware neural nets, hold the promise of discriminating between different mathematical models of sand dune dynamics in a principled way.
4. Climate change can lead to shifts in dune dynamics, including reactivation of dormant dune fields. Careful evaluation and forecasting of such mechanisms is a pressing need in our changing world.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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